Please check that this question paper contains__9____ questions and ___2___ printed pages within first ten

MORNING

[Total No. of Questions: 09] [Total No. of Pages: 2]

10 JUN 2023

Uni. Roll No. ....

Program: B.Tech. (Batch 2018 onward)

Semester: 3rd

Name of Subject: Digital Electronics

Subject Code: ESCS-101

Paper ID: 16012

Scientific calculator is Not Allowed

Time Allowed: 03 Hours **Max. Marks: 60**

NOTE:

1) Parts A and B are compulsory
2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
3) Any missing data may be assumed appropriately

Part – A **[Marks: 02 each]**

Q1.

a) Convert the hexadecimal number $(68.4B)_{16}$ into equivalent octal number.

b) Demonstrate FPGA.

c) Define Minterm and Maxterm. Give example of each.

d) Which is the fastest logic family and why?

e) Change the given expression Y=AB+AC+BC into SOP form.

f) Justify the need of shift registers.

Part – B **[Marks: 04 each]**

Q2. Discuss PLA and PAL.

Q3. Illustrate the working principle of a half adder and draw its truth table.

Q4. Make use of the architecture of a programmable logic device to explain it in detail.

Q5. Justify the need for using signed integers represented in 2's complement in computer programming languages, highlighting the advantages it offers in terms of range and mathematical operations.

Q6. Construct SR Flip-Flop from D-Flip Flop.

Q7. Minimize the given expression using K-map.

$Y = \Sigma m\,(0, 1, 5, 9, 13, 14, 15) + d\,(3, 4, 7, 10, 11)$

Part – C **[Marks: 12 each]**

Q8. List various types of Digital-to-Analog Convertors. Explain R-2R Ladder Digital-to-Analog Converter in detail.

OR

Design Mod-8 Counter using J-K Flip Flops.

Q9. a) Where do we use Excess -3 and Gray codes? **(2marks)**

b) Evaluate $(83)_{10}$ and $(34)_{10}$ in BCD. **(4marks)**

c) Evaluate 108+ 789 in Excess -3 Code. **(6marks)**

OR

a) Simplify the following expressions using Boolean laws: **(6marks)**

$Y = (A+C)\,(A+D)\,(B+C)\,(B+D)$

b) Design a circuit diagram the given expression using NOR gates only:**(6marks)**

$Y = (ABC + B'C')C$

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